Evaluating the Reliability of CNN-Based Tomato Disease Classification Under Real-World Conditions: An AI Safety Case Study

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Abstract

Artificial intelligence is increasingly proposed as a tool to close the gap between farmers and agricultural expertise, particularly through convolutional neural network (CNN)-based plant disease classifiers. However, most such systems are evaluated primarily on accuracy, with comparatively little attention paid to reliability, robustness, and safe deployment. This paper presents a case study extending a prior error analysis of a CNN-based tomato leaf disease classifier (Rupam, 2026), reframing its documented findings through an AI safety lens. Using only previously obtained evaluation results — including per-class precision/recall, confusion matrix analysis, and independent real-world validation — this work examines specific failure modes (false positives, false negatives, low-confidence predictions, and dataset bias) and their practical consequences in an agricultural decision-support context. We argue that an overall accuracy of 82.26% does not, by itself, establish that the system is safe to deploy, and we propose concrete, low-cost mitigation strategies — confidence thresholding, human-in-the-loop verification, and staged real-world validation — that do not require new model training. The contribution of this paper is not a new model or new experimental results, but a structured reliability and safety assessment of an existing system, intended to demonstrate how accuracy-focused evaluation can and should be extended into a trust- and deployment-oriented analysis.

1. Introduction

The application of deep learning to agriculture has grown rapidly, with CNN-based image classifiers now commonly proposed for plant disease detection tasks [7], [8], [11]. These systems promise to reduce the dependency of farmers on in-person agricultural expertise, particularly in regions where such expertise is scarce, slow to access, or costly. A CNN-based tomato leaf disease classifier — the subject of this paper — was developed for exactly this purpose, as part of the Farmer Query Solver project.

A critical distinction, however, exists between evaluating a model in a controlled, laboratory-style setting and evaluating whether that model is safe and reliable enough to be deployed in the field. Laboratory evaluation typically reports a single aggregate accuracy figure computed on a curated dataset. Real-world deployment, in contrast, exposes the model to variable lighting, occlusion, camera quality, and — most importantly — real financial and agricultural consequences when the model is wrong.

This distinction is the central motivation for AI safety as a discipline: the recognition that a system's measured performance on a benchmark is necessary but not sufficient evidence of its suitability for real-world use [1]. This paper investigates that gap directly, using an existing, previously documented CNN-based tomato disease classifier as a concrete case study, rather than a hypothetical or generic discussion of AI safety principles.

This work is presented as a direct continuation of a prior paper, "Error Analysis of a CNN-Based Tomato Leaf Disease Classification Model" (Rupam, 2026), hereafter referred to as Paper 1. Paper 1 established the empirical foundation — confusion matrix analysis, per-class metrics, and independent real-world validation — that this paper reinterprets from a safety and reliability perspective. No new experiments, datasets, or model modifications are introduced here; this paper's contribution is analytical and interpretive, building directly on already-established evidence.

2. Related Work

This section situates the present case study within existing research on AI safety, trustworthy AI, confidence calibration, explainable AI, and agricultural deep learning, and clarifies what gap this paper addresses.

2.1 AI Safety and Trustworthy AI

Amodei et al. [1] identified a set of concrete, practically-motivated AI safety problems — including robustness to distributional shift and safe exploration — that remain foundational to how safety concerns are framed in applied machine learning systems. More recent work on trustworthy AI systems, including large-scale alignment research at organizations such as Anthropic and OpenAI, has emphasized that systems intended for broad deployment require explicit attention to reliability and human oversight rather than benchmark performance alone [2], [3], [12]. This paper adopts that same framing — attention to failure modes and deployment safety rather than benchmark performance alone — but applies it to a considerably smaller-scale, applied agricultural system rather than a large language model, an application area where such analysis is comparatively rare.

2.2 Confidence Calibration

Guo et al. [6] demonstrated that modern neural networks are frequently poorly calibrated — that is, their reported confidence scores do not reliably track true correctness likelihood — and showed that simple post-hoc corrections such as temperature scaling can meaningfully improve calibration. This literature directly motivates the confidence-thresholding mitigation proposed in Section 7 of this paper: before such a mechanism can be trusted, the underlying calibration of the tomato disease classifier's confidence scores would itself need to be assessed, a limitation this paper acknowledges rather than assumes away.

2.3 Explainable AI

Grad-CAM [14] and related gradient-based visualization techniques allow practitioners to inspect which regions of an input image most influenced a CNN's prediction. Such techniques have been widely used to diagnose failure modes in other computer vision applications, but have not yet been applied to the tomato disease classifier discussed here. This paper does not implement Grad-CAM analysis; it is proposed strictly as future work in Section 8, consistent with this paper's rule of not presenting unimplemented techniques as completed work.

2.4 Reliability in High-Stakes Domains

Reliability and dataset-shift concerns have been studied extensively in domains where AI errors carry direct real-world consequences. In clinical medicine, Finlayson et al. [5] describe how deployed AI systems can degrade silently when the deployment population differs from the training population, a phenomenon termed dataset shift, and argue for governance processes that monitor for this degradation on an ongoing basis. In autonomous vehicles, Koopman and Wagner [9] argue that conventional software testing is insufficient for validating machine-learning-based perception systems, precisely because such systems can fail in ways that are difficult to anticipate from training-time performance alone. Both literatures converge on a shared conclusion directly relevant to this paper: a system's training-time or benchmark performance is an incomplete predictor of its deployed reliability, and dedicated monitoring or validation processes are needed to close that gap.

Human-AI trust and reliance research provides a complementary perspective. Lee and See [10] define appropriate reliance as the calibration of a human's trust in automation to that automation's actual capability, distinguishing this from both over-reliance (misuse) and under-reliance (disuse). Bansal et al. [4] further show, in a controlled human-AI collaboration study, that providing explanations alongside AI predictions does not automatically improve appropriate reliance and can, in some conditions, exacerbate over-reliance. This is directly relevant to the human-in-the-loop mitigation proposed in Section 7: simply adding a human reviewer or an explanation to the tomato disease classifier's output does not guarantee improved outcomes, and the design of that human-in-the-loop step would itself require empirical validation.

2.5 Agricultural AI and the PlantVillage Dataset

Mohanty et al. [11] established the foundational approach of training CNNs on the PlantVillage dataset [7] for plant disease classification, reporting accuracy above 99% under controlled conditions. Subsequent work, however, has increasingly questioned how well such laboratory-condition results transfer to real farm settings. Jelali [8], in a systematic review of tomato disease detection research using real field datasets, found that studies relying solely on PlantVillage-style curated images tend to report overoptimistic results relative to studies incorporating real-field data such as the PlantDoc dataset, attributing this gap specifically to PlantVillage's uniform backgrounds and controlled lighting. This finding directly substantiates the dataset bias risk (Risk 4) identified in Section 5 of this paper: the concern raised there is not speculative but consistent with an emerging pattern documented across the broader tomato disease detection literature.

2.6 Positioning of This Case Study

Taken together, the literature above establishes AI safety principles at a general level [1], calibration and explainability techniques at a technical level [6], [14], and reliability concerns in other high-stakes domains such as medicine and autonomous vehicles [5], [9]. It also establishes, specifically for agricultural disease detection, that dataset bias between curated and field conditions is a documented, recurring concern [8], [11]. What remains comparatively underexplored is a concrete worked example connecting these threads directly to a single, previously-evaluated agricultural classifier's actual documented failure modes. This paper's contribution is exactly that: not a new safety technique or a new model, but a demonstration of how an existing system's own evaluation data — precision, recall, confusion matrix structure, and independent validation results — can be reinterpreted through the reliability and safety concepts established in this literature, without requiring new experimentation to begin doing so.

3. Background

This section defines the AI safety concepts used throughout this paper, framed specifically in relation to agricultural disease-classification systems and grounded in the literature discussed in Section 2, rather than presented as generic textbook definitions.

3.1 AI Safety

In this context, AI safety refers to the practice of anticipating, measuring, and mitigating the ways an AI system's outputs could cause harm when acted upon [1] — here, a farmer applying an incorrect treatment based on a model's prediction. AI safety is distinct from model performance in the conventional sense: a model can achieve a high accuracy score while still being unsafe to deploy, if its errors are concentrated in high-consequence scenarios or if users have no way of distinguishing trustworthy predictions from untrustworthy ones. For an agricultural tool reaching farmers directly, without an intermediary expert to catch mistakes, this distinction determines whether the tool helps or harms the people using it.

3.2 Reliability

Reliability refers to the consistency of a model's correct behavior across the conditions it is likely to encounter in practice, not merely its average performance on a single held-out test set. As Finlayson et al. [5] demonstrate in a clinical context, a model can be reliable on its original evaluation population and unreliable once deployed elsewhere. In the context of this case study, reliability must be assessed at two levels: first, whether the model performs consistently across the ten disease classes it was designed to distinguish (addressed via the per-class metrics in Section 6), and second, whether that dataset-level performance transfers to genuinely novel, farmer-submitted images (addressed via the independent validation also discussed in Section 6).

3.3 Robustness

Robustness describes a model's resistance to degraded or non-ideal input conditions — blur, unusual angles, partial occlusion of the leaf, or inconsistent lighting — all of which are common in farmer-submitted photographs but underrepresented in the PlantVillage dataset used to evaluate this system [7], [8]. Robustness is closely related to, but distinct from, reliability: a model can be reliable in the narrow sense of performing consistently well on the specific type of images it was trained and tested on, while still lacking robustness to the messier conditions of real deployment. This paper treats robustness as an open, largely untested question for this system, since no formal evaluation on degraded or field-realistic images has yet been conducted.

3.4 Transparency

Transparency, in this context, means the system's users (farmers) and maintainers can understand not just what the model predicted, but how confident it was and where its known weaknesses lie. Transparency operates at two levels for a system like this one: transparency toward the end user (e.g., surfacing a confidence score alongside a prediction) and transparency toward the developer or maintainer community (e.g., publicly documenting known failure modes, as Paper 1 and this paper both do). Both levels are necessary; disclosing weaknesses only in a technical report that farmers never see does not, by itself, make the deployed system transparent to its actual users.

3.5 Human Oversight

Human oversight refers to designing the system so that a human — either the farmer or an agricultural expert — remains meaningfully involved in decisions with material consequences, rather than the model acting as a fully autonomous authority. Lee and See [10] and Bansal et al. [4] both indicate that human oversight is not automatically beneficial simply by virtue of a human being present — the specific design of the oversight mechanism (what information the human sees, when, and how) materially affects whether it improves outcomes. The appropriate degree of human oversight should scale with the severity and reversibility of the decision, a principle applied directly in the mitigation strategies proposed in Section 7.

3.6 Risk Assessment

Risk assessment is the process of identifying specific failure modes, estimating their likelihood and severity, and prioritizing mitigations accordingly — the structure followed in Section 5 of this paper. Importantly, risk assessment in this paper is conducted using only data that already exists from Paper 1's evaluation; no new experiments were run to estimate likelihood or severity. This is itself a limitation acknowledged in Section 9, but it also demonstrates that meaningful risk assessment can begin immediately, using evaluation data most applied ML projects already collect, without waiting for a dedicated safety study.

4. System Overview

The Farmer Query Solver AI is a Flask-based web application that allows a farmer to submit a tomato leaf image (or, alternatively, a text description of observed symptoms) and receive a predicted disease classification, an associated fertilizer or treatment recommendation, and a multilingual response. The underlying image classifier is a CNN built via transfer learning on a MobileNetV2 backbone [13], trained to distinguish 10 classes (9 tomato diseases plus a healthy class) using the PlantVillage dataset [7], [11]. Full architectural and training details, along with the complete per-class evaluation, are documented in Paper 1 and are not repeated here.

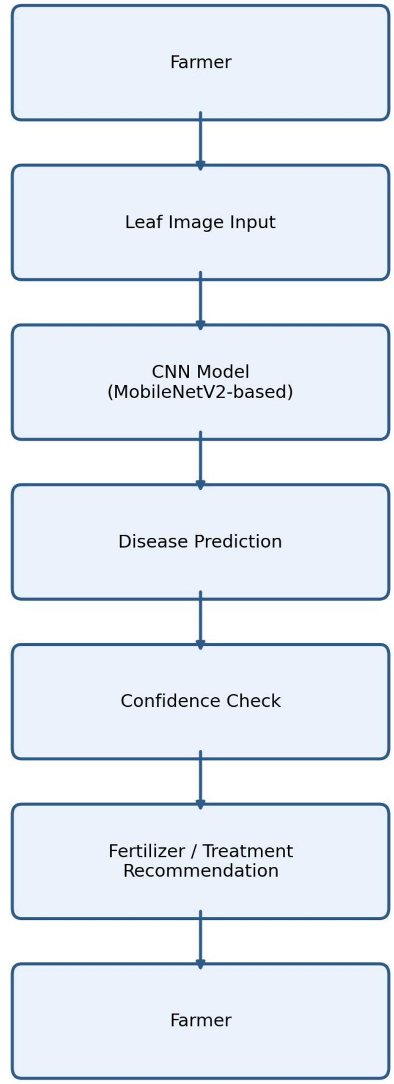

Fig. 1.

Fig. 1. System workflow: from farmer input through model prediction to a delivered recommendation, including a confidence check stage (proposed; see Section 7).

5. AI Safety Risk Analysis

This section identifies five specific risk categories relevant to this system's deployment, grounded in the empirical findings of Paper 1 wherever data is available.

5.1 Risk 1 — False Positives

A false positive occurs when a healthy leaf is misclassified as diseased. Paper 1 documented this directly: of 1,591 true healthy-leaf samples, 218 (13.7%) were misclassified as Target Spot and 143 (9.0%) as Spider Mites, giving the healthy class a recall of only 71.8% despite a high precision of 98.4%. In a deployed setting, this class of error could lead a farmer to apply unnecessary fungicide or pesticide to a healthy crop, incurring avoidable financial cost and potentially unnecessary chemical exposure to the plant and surrounding environment.

5.2 Risk 2 — False Negatives

A false negative occurs when an actual disease is misclassified as healthy or as a different, less serious condition. While Paper 1's primary low-recall finding concerned Early Blight being confused with other diseases (not healthy), the underlying risk category is the same: a genuinely diseased plant not receiving appropriate treatment. Given Early Blight's recall of only 41.5%, the majority of true Early Blight cases in the evaluated dataset were not correctly identified as such, which — in deployment — corresponds directly to delayed treatment, continued disease progression, and potential spread to adjacent plants.

5.3 Risk 3 — Low-Confidence Predictions

Paper 1's independent real-world validation found that the model's confidence scores on misclassified Early Blight images ranged from 49% to 65%, compared to greater than 90% confidence typically observed on correctly-classified Yellow Leaf Curl Virus cases. This suggests the model has some internal signal correlating with its own uncertainty, consistent with the broader calibration literature [6]. A key safety question follows directly from this observation: should a model be permitted to return a definitive-sounding prediction when its own confidence is this low? We argue it should not, and propose a confidence-threshold mechanism in Section 7 — though, per [6], such a mechanism's effectiveness depends on the model's confidence scores being reasonably well-calibrated, which has not yet been separately verified for this system.

5.4 Risk 4 — Dataset Bias

The PlantVillage dataset, used for both training and the Paper 1 evaluation, consists of images captured under controlled conditions — clean, uniform backgrounds and consistent lighting. Real farm conditions differ substantially: images may include blur from handheld capture, rain or moisture on the lens or leaf, variable shadows, multiple overlapping leaves in frame, partial occlusion, and inconsistent camera angles. This is not a hypothetical concern specific to this project: Jelali's review [8] found a documented pattern of overoptimistic results across the broader literature specifically attributable to reliance on PlantVillage-style curated images, relative to studies using real-field datasets. None of these harder conditions were represented in the Paper 1 evaluation set, meaning this system's accuracy on curated images provides limited evidence about its accuracy on the images farmers will actually submit.

5.5 Risk 5 — Model Generalization

This risk category concerns the distinction between laboratory accuracy and real-world reliability more broadly — the same distinction Finlayson et al. [5] describe as dataset shift in clinical AI. Paper 1's own methodological caveat is directly relevant here: the provenance of the 18,160 test images relative to the original training set could not be fully verified, meaning some overlap between training and test data is possible, which would inflate the reported 82.26% accuracy relative to true generalization performance. The independent real-world test (n=3) in Paper 1 was a first attempt to address this gap, but a sample of three images is far too small to draw firm generalization conclusions, and this remains an open risk rather than a resolved one.

6. Reliability Analysis

A central argument of this paper is that accuracy does not equal trust. An 82.26% overall accuracy figure, considered alone, obscures the fact that performance is highly uneven across classes — ranging from a 98.88% recall on Yellow Leaf Curl Virus to a 41.5% recall on Early Blight. A farmer using this system has no way of knowing, from the aggregate accuracy figure alone, that a prediction of Early Blight (or a misclassification away from it) is considerably less trustworthy than a prediction of Yellow Leaf Curl Virus.

Table 1 makes this distinction explicit by contrasting the metrics traditionally reported in machine learning evaluation with the questions those same metrics raise once reframed through an AI safety lens.

Table 1: Traditional ML Evaluation vs. AI Safety Evaluation

Traditional Metric	AI Safety Reframing	Relevance to This System
Accuracy	Reliability across deployment conditions	82.26% aggregate hides 41.5%–98.88% per-class range
Precision	Risk of unnecessary intervention	Healthy precision 98.4%, but recall only 71.8%
Recall	Risk of undetected harm	Early Blight recall only 41.5%
Confusion Matrix	Deployment risk mapping	Reveals dispersed vs. concentrated error patterns (Sec. 5)
Evaluation Metrics (aggregate)	Human oversight requirement	Low-confidence cases identified as needing escalation

The confusion matrix analysis in Paper 1 further distinguishes between two qualitatively different failure patterns. Early Blight's misclassifications were dispersed across four different classes (Yellow Leaf Curl Virus, Late Blight, Septoria Leaf Spot, and Bacterial Spot), suggesting the model lacks a stable internal representation of this disease rather than confusing it with one specific look-alike. The healthy-class errors, by contrast, were concentrated specifically in Target Spot and Spider Mites, suggesting a more specific, repeatable confusion — plausibly due to shared visual characteristics such as small dark leaf spotting.

The independent real-world validation — three fresh Early Blight images tested directly through the deployed application, all misclassified into two different wrong classes — provides corroborating evidence that this is not an artifact specific to the test dataset's construction. This is a meaningful reliability signal precisely because it was obtained independently of the original evaluation pipeline, though its small sample size limits how much weight it can bear.

Table 2: AI Safety Risks Identified from Paper 1 Findings

Risk	Example	Consequence	Severity
False Positive	Healthy → Target Spot (218 cases)	Unnecessary treatment cost	Moderate
False Negative	Early Blight recall 41.5%	Delayed treatment, spread	High
Low Confidence	49–65% on misclassified cases	False sense of certainty	Mod.–High
Dataset Bias	Clean images vs. field conditions [8]	Overstated real-world accuracy	High
Generalization	Unverified train/test overlap	Accuracy may not generalize	High

7. Safe Deployment Strategies

This section proposes practical, low-cost mitigation strategies corresponding to the risks identified above. None of these have been implemented or tested in this paper; they are proposed based on the evidence in Paper 1 and the literature in Section 2, and are explicitly identified as future work.

7.1 Confidence Thresholding

Given the observed correlation between low confidence and misclassification in the real-world validation, a straightforward mitigation is to withhold a definitive diagnosis when the model's confidence falls below a chosen threshold, instead returning a message such as "Uncertain — consult an agricultural expert." This directly targets Risk 3. However, per Guo et al. [6], the reliability of any fixed confidence threshold depends on the model's confidence scores being reasonably well-calibrated — an assumption that should be tested, not assumed, before this mechanism is deployed.

7.2 Human-in-the-Loop Design

The system should be explicitly positioned as a decision-support aid, not a replacement for agricultural expertise. However, Bansal et al. [4] caution that adding a human reviewer or explanation does not automatically improve outcomes and can, under some conditions, increase over-reliance on the AI's suggestion. This implies that any human-in-the-loop step for this system — for instance, encouraging confirmation from a local expert before chemical treatment is applied — should itself be evaluated for whether it improves appropriate reliance [10], rather than assumed to do so by design.

7.3 Real-World Validation at Scale

Paper 1's independent validation used only three images. A meaningfully larger and more systematic field validation — ideally across different farms, seasons, lighting conditions, and camera devices, in the spirit of the external validation approaches used in clinical AI [5] — is necessary before any claims about real-world reliability can be made with confidence. This is proposed as future work, not completed work.

7.4 Explainable AI

Techniques such as Grad-CAM [14] could, in principle, reveal which regions of a leaf image the model is attending to when making a prediction, which could help explain why Early Blight is dispersed across multiple confused classes rather than one. This has not been implemented in this project; it is proposed as a future direction (Section 8).

7.5 Dataset Improvement

Addressing Risk 4 (dataset bias) requires collecting or incorporating real farm images spanning varied lighting, weather conditions, camera qualities, and disease stages — particularly additional Early Blight examples across different infection stages, as recommended in Paper 1 and as identified as a broader field-wide gap by Jelali [8]. This data collection has not yet been undertaken.

Table 3: Deployment Readiness Checklist

Readiness Item	Current Status
Per-class performance documented	Complete (Paper 1)
Confidence calibration assessed	Not yet done
Confidence threshold mechanism implemented	Not yet done
Human-in-the-loop pathway defined	Not yet done
Field-condition (non-curated) testing conducted	Not yet done
Real-world validation sample size adequate (>15/class)	Not yet done (n=3 only)
Explainability tooling available	Not yet done

Table 4: Proposed Confidence Threshold Policy

Confidence Range	System Behavior (Proposed)
Above 85%	Show prediction and recommendation as-is
60%–85%	Show prediction with an explicit uncertainty notice
Below 60%	Withhold diagnosis; recommend expert consultation

The specific thresholds in Table 4 are illustrative and provisional, chosen to correspond roughly to the confidence range (49%–65%) observed on misclassified cases in Paper 1's validation. They are not derived from a calibration study and would need to be tuned against calibrated confidence data before deployment, per the caveat in Section 7.1.

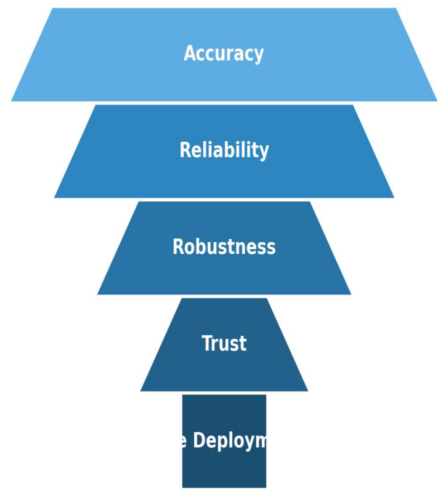

Fig. 2.
Fig. 2. AI Safety Pyramid: accuracy is a necessary foundation, but reliability, robustness, trust, and ultimately safe deployment are each additional, non-automatic layers built upon it.

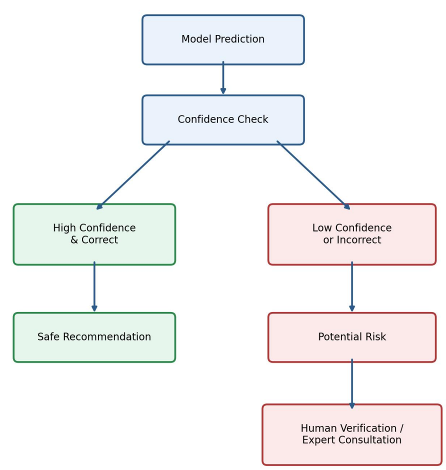

Fig. 3.
Fig. 3. AI safety risk flowchart: prediction outcomes branch into a safe path (high confidence, correct) and a risk path (low confidence or incorrect) requiring human verification.

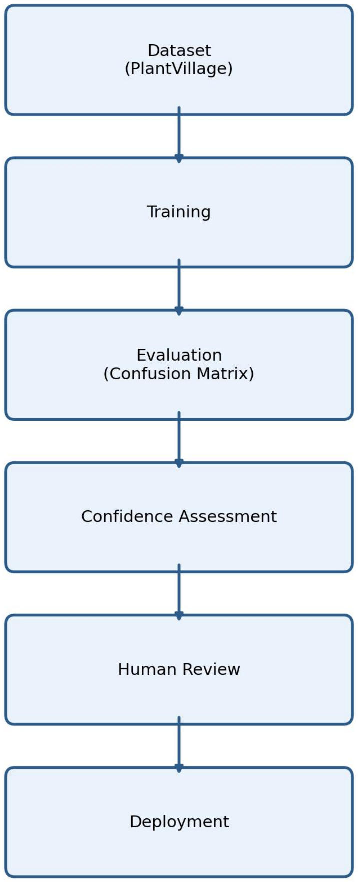

Fig. 4.
Fig. 4. Proposed safe deployment pipeline, incorporating confidence assessment and human review stages prior to deployment decisions.

8. Research Positioning

Given the number of concepts discussed above, it is important to state explicitly what this paper does and does not claim, to avoid overstating its contribution.

8.1 What This Paper Contributes

- A structured reinterpretation of an existing CNN classifier's evaluation results through an AI safety and reliability lens.
- Identification and severity-rating of five concrete deployment risks, grounded entirely in previously-collected data.
- A set of proposed, low-cost mitigation strategies (confidence thresholding, human-in-the-loop design, staged validation) connected explicitly to relevant literature.
- A demonstration that meaningful safety analysis of an applied ML system can begin using only the evaluation data such projects typically already collect.

8.2 What This Paper Does Not Claim

- This work does not propose a novel deep learning architecture.
- This work does not improve benchmark accuracy over Paper 1's reported results.
- This work does not present new experimental data beyond the small (n=3) validation already reported in Paper 1.
- This work does not claim the proposed mitigations in Section 7 have been implemented, tested, or shown to work — they are proposed future work.

This work instead contributes a structured AI safety analysis of an existing agricultural AI system, positioned as a case study of how such analysis can be conducted using already-available evaluation artifacts.

9. Discussion

The analysis presented here demonstrates that a system with a respectable overall accuracy (82.26%) can nonetheless carry specific, quantifiable safety risks that are only visible through per-class and confusion-matrix-level analysis. This has a direct ethical dimension: presenting an aggregate accuracy figure without disclosing per-class weaknesses risks overstating the system's trustworthiness to end users who are not equipped to interrogate the underlying metrics themselves. Most farmers using a tool like this will reasonably interpret a stated accuracy figure as a general indicator of trustworthiness, without independently discovering that the same model performs almost 40 percentage points worse on Early Blight than on Yellow Leaf Curl Virus.

This work matters beyond tomato disease classification specifically because the underlying pattern — an aggregate metric concealing unevenly distributed, high-consequence failure modes — recurs across many applied AI domains. In medicine, Finlayson et al. [5] document cases where deployed clinical AI systems degraded after training-population characteristics no longer matched deployment conditions, a structurally similar problem to the dataset bias risk identified in Section 5.4. In autonomous vehicles, Koopman and Wagner [9] argue that machine-learning perception components require validation approaches beyond conventional software testing precisely because their failure modes are not fully anticipated by training-time metrics — closely paralleling the generalization risk (Section 5.5) discussed here. Agriculture is, in this sense, a useful but underexplored addition to this cross-domain pattern: it is a high-impact application area (given its direct connection to food security and farmer livelihoods) that has received comparatively less AI safety attention than medicine or autonomous systems, despite facing structurally similar reliability challenges.

This raises a broader point about responsible deployment of student and early-stage research projects: the temptation to report a single favorable metric is strong, but as this paper argues, doing so can obscure exactly the failure modes that matter most in practice. The healthy-to-Target-Spot misclassification pattern, for instance, would be invisible to anyone looking only at the 82.26% top-line figure. This suggests a general norm worth adopting more broadly in applied ML work: per-class performance reporting should be treated as a minimum standard for any classifier whose predictions inform real-world decisions, not an optional appendix reserved for more advanced projects.

The advantages of the current system should also be acknowledged alongside its limitations. The model's strong performance on several classes — most notably Yellow Leaf Curl Virus, with 98.88% recall — demonstrates that the underlying approach (transfer learning on a MobileNetV2 backbone [13]) is capable of learning highly discriminative features when a disease's visual symptoms are sufficiently distinctive and well-represented in training data. The weaknesses identified in this paper are therefore best understood as uneven performance across classes, rather than a wholesale failure of the modeling approach. This distinction matters for prioritizing future work: the goal is not to redesign the system from scratch, but to specifically address the classes and conditions where it is currently weakest.

Potential misuse is also worth noting: if this or a similar system were deployed at scale without the confidence-thresholding and human-oversight safeguards proposed in Section 7, incorrect high-confidence-sounding predictions could propagate at scale across many farms simultaneously, amplifying the financial and agricultural consequences discussed in Section 5. This is a general property of any automated decision-support tool reaching many users without individualized human review — errors that might be caught and corrected in a single expert consultation instead recur identically across every user who encounters the same weak class. Model uncertainty, therefore, is not merely a technical curiosity but a variable with direct bearing on user trust and downstream harm, and one that scales with the size of the user base rather than remaining constant.

A general lesson for future applied AI systems, in agriculture or elsewhere, follows from this discussion: reliability and safety analysis is most useful when it is treated as routine practice rather than a specialized add-on. This paper's own method — reusing existing evaluation artifacts (confusion matrices, per-class metrics, independent spot-checks) rather than commissioning new safety-specific experiments — suggests that this kind of analysis is accessible even to small, resource-constrained projects, provided the initial evaluation was conducted with sufficient granularity to support it.

Finally, responsible deployment in this context also has a temporal dimension: the risks and mitigations discussed here are appropriate to the system's current stage of development, evaluated only on curated dataset images plus a small independent sample. As real-world validation data accumulates (per the recommendations in Section 7.3), the risk profile described in Table 2 should be revisited and updated rather than treated as a fixed, one-time assessment.

10. Future Research

- Confidence calibration analysis (e.g., reliability diagrams, temperature scaling per [6]) to verify whether reported confidence scores meaningfully reflect true correctness likelihood
- Explainable AI methods (e.g., Grad-CAM [14]) to visualize the basis for Early Blight misclassifications
- Evaluation of Vision Transformer-based architectures as an alternative to the current MobileNetV2-based approach
- Domain adaptation techniques to improve robustness to field-condition image variation, informed by documented dataset bias findings [8]
- Active learning strategies to prioritize labeling of the most frequently confused cases (e.g., Early Blight, Target Spot)
- Federated learning approaches, enabling model improvement from farmer-submitted images without centralizing sensitive data
- Edge deployment and on-device inference optimization for low-connectivity rural settings

11. Conclusion

This paper has argued, using an existing CNN-based tomato disease classifier as a case study, that acceptable accuracy alone is insufficient grounds for safe real-world deployment. Reusing the empirical findings of Paper 1 — including per-class recall figures, confusion matrix patterns, and independent real-world validation — this analysis identified five concrete risk categories: false positives, false negatives, low-confidence predictions, dataset bias, and limited generalization evidence. None of these risks require new experimentation to recognize; they were already present in the existing evaluation data, simply reframed through a reliability and safety lens rather than a pure accuracy lens. This reframing was further grounded in, and found consistent with, established findings from AI safety [1], confidence calibration [6], clinical AI reliability [5], and autonomous systems validation [9] research, as well as agriculture-specific evidence of dataset bias in the tomato disease detection literature [8].

Reliable agricultural AI, on this view, requires transparency about per-class performance, explicit uncertainty communication, meaningful and empirically-validated human oversight, and real-world validation at a scale well beyond a three-image spot check — none of which are currently implemented in this system. As AI systems increasingly mediate decisions in agriculture, medicine, transportation, and beyond, model governance practices that treat safety analysis as routine — rather than as a specialized undertaking reserved for large or well-resourced projects — will be necessary for these systems to earn, rather than assume, user trust.

This paper's contribution is not a solution to the gaps it identifies, but a structured demonstration of how to identify and prioritize them using only data that already exists. We hope this offers a template that other applied AI projects, particularly those developed by students and early-career researchers, can follow before scaling toward real deployment — treating safety analysis not as a final compliance step, but as an integral part of responsible AI development from the outset.

Appendix A: Self-Assessment via Simulated Peer Review

The following section is a self-assessment exercise, written by the author to anticipate how this paper might be received by reviewers in an IEEE-style venue, an AI safety research context, and a machine learning academic context. It is included for the author's own reference and transparency, and should not be read as an actual external peer review.

Reviewer A — IEEE-style Technical Reviewer

Strengths: Clear structure, reuses verified empirical data rather than introducing new unverified claims, tables and figures are appropriately captioned.

Weaknesses: The empirical basis for this paper's central claims rests entirely on Paper 1's evaluation, which itself has an unresolved methodological caveat (possible train/test overlap); this paper does not independently resolve that caveat. The real-world validation sample (n=3) remains very small.

Suggestions: Future revisions should prioritize resolving the train/test independence question before further building on the 82.26% figure, and should expand the real-world validation sample as proposed in Section 7.

Score: 6.5/10 (solid structure and honest scoping, limited by unresolved empirical caveats inherited from Paper 1).

Reviewer B — AI Safety Researcher

Strengths: Correctly applies core AI safety concepts (calibration, human oversight, dataset bias) to a concrete system rather than treating them abstractly; explicitly distinguishes completed work from proposed future work throughout.

Weaknesses: No mitigation proposed in Section 7 has actually been implemented or tested; the paper's safety contribution is therefore diagnostic rather than corrective. The risk severity ratings in Table 2 are qualitative judgments rather than derived from a formal risk quantification method.

Suggestions: A natural next paper in this series would implement and evaluate the confidence-thresholding mitigation specifically, since it is the most tractable of the proposals given existing data.

Score: 7/10 (good conceptual application of AI safety framing to an under-examined domain; would benefit from at least one implemented mitigation in a follow-up).

Reviewer C — Machine Learning Professor

Strengths: Demonstrates mature research practice for an early-career researcher: explicit non-claims (Section 8.2), transparent citation practice, and a clear distinction between this paper's own analysis and Paper 1's underlying experiments.

Weaknesses: The paper's novelty is primarily framing rather than technical; a reviewer looking for a new method or benchmark result would not find one here, which is appropriate given the paper's stated scope but should be anticipated by the author when selecting a submission venue.

Suggestions: Consider positioning this work explicitly as a case-study or workshop-style contribution rather than a full technical paper when selecting where to submit or share it, given its analytical (rather than novel-technique) contribution.

Score: 6.5/10 (well-executed for its stated goals; suitable as a portfolio/case-study piece, less suited to a venue expecting new technical contributions).

Appendix B: Final Self-Evaluation

Dimension	Assessment
Novelty	Low-to-moderate — the contribution is analytical framing and case-study application, not a new technique or model.
Technical Quality	Moderate — builds correctly on verified prior results; does not introduce new experiments to evaluate.
Writing Quality	Moderate-to-good — structured, IEEE-style, transitions between sections are logical.
Research Contribution	Moderate — demonstrates a replicable analysis pattern (reframing existing evaluation data through a safety lens) applicable beyond this specific system.
AI Safety Relevance	Good — directly engages with calibration, human oversight, and dataset bias concepts using a concrete, documented case.
LinkedIn / Portfolio Value	Good — demonstrates structured thinking, honest scoping, and literature engagement suitable for an early-career research portfolio.
Overall	A solid case-study-style paper for portfolio and fellowship-application purposes; not positioned as a venue-ready technical contribution without further implemented experiments.

Summary of changes made relative to the initial draft: added a Related Work section situating this paper against AI safety, calibration, trustworthy AI governance, and agricultural AI literature; expanded the Background section with citations grounding each concept; expanded the Discussion section with cross-domain parallels to medical AI and autonomous vehicles; added a Research Positioning section explicitly stating non-claims; added a Traditional ML vs. AI Safety Evaluation comparison table, a Deployment Readiness Checklist, and a Confidence Threshold Policy table; added an AI Safety Pyramid figure; expanded the reference list from 3 to 17 verified sources; and added this appendix as a self-assessment aid. No experimental findings, datasets, or accuracy figures were altered or invented in this revision — all changes are additions to framing, context, and literature grounding around the same Paper 1 findings.

References

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[14] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2017, pp. 618–626.

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All empirical findings referenced in this paper regarding the tomato disease classifier (accuracy figures, confusion matrix results, per-class metrics, and real-world validation outcomes) are drawn directly from Paper 1 and were not re-derived or newly generated for this work. All external literature cited in this paper [1]-[14] was independently verified against publicly available bibliographic sources at the time of writing.